

Type of article: Original

Analysis of Measles Transmission Dynamics Using the Unscented Kalman Filter

Muhammad Muafi ¹, ORCID : <https://orcid.org/164221092>

Herman Effendi ², ORCID : <https://orcid.org/164221056>

Andreas Hendra Herwanto³, ORCID: <https://orcid.org/164221064>

Pradipta Deska Pryanda⁴, ORCID: <https://orcid.org/164221108>

^{1, 2, 3, 4} Universitas Airlangga, Engineering Department. Surabaya, Indonesia.

ABSTRACT

Introduction: Measles remained a highly contagious infectious disease, requiring sustained high vaccination coverage to prevent outbreaks. In Indonesia, measles incidence exhibited substantial temporal variability influenced by changes in immunization coverage and surveillance conditions. Traditional deterministic models often failed to capture the non-linear and noisy characteristics of epidemiological data, motivating the application of state-space filtering approaches.

Methods: This study analyzed annual measles surveillance data in Indonesia from 2004 to 2024 obtained from the World Health Organization, including reported measles cases and first- and second-dose measles-containing vaccine coverage. An Unscented Kalman Filter was implemented within a non-linear state-space framework to estimate latent measles incidence dynamics. Reported cases were log-transformed and treated as noisy observations, while vaccination coverage variables were incorporated as exogenous inputs. Model performance was evaluated using RMSE, NRMSE, and MAPE.

Results: The Unscented Kalman Filter effectively captured the underlying dynamics of measles incidence and produced smoothed estimates that closely followed observed trends, including the sharp decline around 2020-2021 and the resurgence in subsequent years. The model achieved strong performance, with an RMSE of 763.79 cases, NRMSE of 0.0265, and MAPE of 5.41%. The estimated parameters indicated a positive latent trend in recent years and a more consistent protective effect of second-dose vaccination coverage.

Conclusions: The findings demonstrated that the Unscented Kalman Filter provided a robust and interpretable approach for modeling measles surveillance data under non-linear and noisy conditions, with potential utility for infectious disease monitoring and public health decision-making.

Keywords: Measles; Unscented Kalman Filter; State-space model; Vaccination coverage; Epidemiological surveillance.

1. INTRODUCTION

Measles represents one of the most highly contagious pathologies in human epidemiology, with a natural Basic Reproduction Number (R_0) estimated between 12 and 18 (1). This level of transmissibility significantly exceeds that of diseases such as diphtheria or early COVID-19 variants, thereby requiring extremely high immunization coverage, above 95%, to achieve herd immunity. In the Indonesian context, measles transmission dynamics are highly sensitive to fluctuations in routine immunization coverage, both for the first dose and the booster (second dose). Declines in vaccination coverage observed during the COVID-19 pandemic have increased the risk of outbreaks in several provinces (2).

The primary challenge in mitigating this risk lies in the limitations of conventional epidemiological modeling approaches. Classical deterministic compartmental models, such as the SIR or SEIR frameworks, often rely on rigid assumptions, including constant transmission parameters and homogeneous mixing, which may not reflect real-world conditions (3). In standard formulations, the transmission rate (β) is frequently treated as time-invariant; however, in practice, this parameter evolves dynamically due to changes in population immunity, public health interventions, behavioral responses, and external shocks. Moreover,

epidemiological surveillance data are typically affected by substantial measurement noise, reporting delays, and underreporting, which further reduce the reliability of deterministic estimates (3,4).

To address these limitations, stochastic approaches based on filtering theory have been proposed as a more flexible alternative for infectious disease modeling. Ghani (2025) introduced the use of the Extended Kalman Filter (EKF) to estimate transmission dynamics in a diphtheria model, demonstrating that recursive Bayesian filtering can effectively accommodate data uncertainty and provide dynamically updated parameter estimates. The EKF operates by linearizing nonlinear state equations through a first-order Taylor series expansion around the current state estimate, enabling sequential correction of model predictions as new observations become available. From a data science and control systems perspective, this approach can be interpreted as a form of Bayesian state estimation for dynamic systems, offering advantages such as nowcasting capabilities and robustness to noisy observations (4).

However, despite its advantages, the EKF relies heavily on local linearization via Jacobian matrices, which can introduce approximation errors when the underlying system exhibits strong nonlinearity or rapid structural changes (5). In the context of measles transmission, where incidence dynamics are influenced by complex and nonlinear interactions between vaccination coverage, population susceptibility, and external disruptions, such linearization may limit estimation accuracy and numerical stability. To overcome these limitations, this study adopts the Unscented Kalman Filter (UKF), which avoids explicit linearization by propagating a set of deterministically chosen sigma points through the nonlinear system dynamics. The Unscented Transform employed by the UKF enables more accurate estimation of posterior means and covariances, particularly in highly nonlinear and noise-dominated epidemiological settings (6).

Accordingly, this research extends the stochastic filtering framework proposed by Ghani (2025) to the analysis of measles dynamics in Indonesia using annual surveillance data from 2004 to 2024. The novelty of this study lies in the formulation of a state-space model in which measles incidence is treated as a latent process, while key epidemiological drivers, including vaccination coverage, are incorporated as exogenous components. By allowing the underlying transmission dynamics to evolve over time rather than assuming constant parameters, the proposed UKF-based approach enables adaptive estimation of measles incidence and provides a more realistic representation of long-term transmission patterns. This framework offers the potential to support early detection of epidemiological shifts and to inform public health decision-making more effectively than traditional deterministic models.

II. METHODS

A. Data Source

The data used in this study were obtained from the World Health Organization (WHO) and consist of annual measles surveillance records covering the period from 2004 to 2024. The dataset includes reported measles cases as the primary epidemiological variable, along with measles-containing vaccine coverage for the first dose and second dose as supporting explanatory variables. All data are reported at an annual temporal scale, which is suitable for examining long-term trends and structural changes in measles incidence. The inclusion of vaccination coverage data allows the analysis to account for the potential impact of immunization programs on measles dynamics within the Unscented Kalman Filter framework. A detailed description of the variables used in this study, including their definitions and roles in the analysis, is presented in Table 1.

Table 1. Description of Research Variables.

Variable Name	Description
Year	Observation year (2004-2024), representing the temporal index of the dataset
Measles Reported Cases	Annual number of reported measles cases used as the observed epidemiological outcome

Measles Vaccine Coverage (1st Dose)	Percentage of the population receiving the first dose of measles-containing vaccine
Measles Vaccine Coverage (2nd Dose)	Percentage of the population receiving the second dose of measles-containing vaccine

B. Data Preprocessing

Data preprocessing was conducted to ensure data quality, numerical stability, and compatibility with the Unscented Kalman Filter framework. This stage focuses on preparing the raw measles surveillance data into a structured and reliable form for time-series modeling, while minimizing potential sources of bias and computational instability.

B.1. Data Cleaning

Data cleaning was performed as an initial step to ensure temporal consistency and numerical validity of the dataset. The data were first sorted chronologically from the earliest to the most recent year to preserve the correct temporal order required for recursive filtering. The reported measles cases, originally stored as text values due to the presence of thousands separators, were converted into numerical format by removing non-numeric characters and casting the values to floating-point numbers. This conversion enables mathematical operations and model estimation within the UKF framework.

B.2. Missing Value

The dataset was examined for missing and invalid values prior to modeling. Observations containing missing values (NaN) or epidemiologically invalid entries, such as negative case counts, were excluded from subsequent analysis steps. This procedure ensures that the filtering process operates only on valid observations and prevents the propagation of errors during state estimation and model evaluation. Missing values were handled conservatively by omission rather than imputation, given the relatively small dataset size and annual temporal resolution.

B.3 Centering of Exogenous Variables (Vaccine Coverage)

To improve numerical stability and interpretability of the model parameters, the vaccination coverage variables for the first and second measles vaccine doses were mean-centered prior to analysis. This centering process involves subtracting the average coverage value from each observation, resulting in exogenous inputs with zero mean. Centering reduces scale-related issues and helps prevent collinearity between the intercept-like components of the state vector and vaccination effect parameters. This preprocessing step supports more stable estimation of vaccination effects within the state-space model.

C. Metrics Evaluation

Model performance in this study was evaluated using three commonly used error metrics, namely Root Mean Squared Error (RMSE), Normalized Root Mean Squared Error (NRMSE), and Mean Absolute Percentage Error (MAPE). These metrics were selected to provide a comprehensive assessment of the accuracy of the Unscented Kalman Filter in estimating measles incidence by quantifying absolute error, relative error, and scale-independent performance.

C.1. Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) measures the average magnitude of the squared differences between the observed values and the model estimates. RMSE penalizes large errors more heavily due to the squaring operation, making it particularly sensitive to significant deviations between observed measles cases and filtered estimates (13). RMSE is defined as :

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

where,

y_i : represents the observed measles cases,
 $\hat{y}_i$: represents the UKF-estimated measles cases
 n : is the number of observations.

In this study, RMSE is expressed in case units, allowing direct interpretation of the average estimation error in terms of the number of measles cases.

C.2. Normalized Root Mean Squared Error (NRMSE)

Normalized Root Mean Squared Error (NRMSE) is a scale-independent version of RMSE that facilitates comparison across datasets with different magnitudes. NRMSE was computed by normalizing RMSE using the range of the observed measles cases (7). The NRMSE is calculated as :

$$NRMSE = \frac{RMSE}{y_{max} - y_{min}}$$

where,

y_{max} & y_{min} : denote the maximum and minimum observed measles cases, respectively.

By normalizing the error, NRMSE provides a relative measure of model performance, indicating how large the estimation error is compared to the overall variability of the data. Lower NRMSE values indicate better model performance.

C.3. Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error (MAPE) measures the average relative error between observed and estimated values, expressed as a percentage. MAPE is particularly useful for interpreting model accuracy in relative terms, especially when dealing with epidemiological data where case counts may vary substantially over time (7). MAPE is defined as :

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

where all variables retain their previous definitions.

To avoid division by zero, a small adjustment was applied when observed case values were extremely small. In this study, MAPE provides an intuitive interpretation of the average percentage deviation between UKF estimates and reported measles cases.

III. DEVELOPMENT

This chapter details the mathematical formulation of the estimation algorithm employed in this study. To address the inherent non-linearities in the Measles transmission model—specifically the interactions between susceptible and infected populations ($S(t)I(t)$), the Unscented Kalman Filter (UKF) is utilized. Unlike the Extended Kalman Filter (EKF) which relies on first-order linearization via Jacobian matrices, the UKF employs a deterministic sampling approach known as the Unscented Transform to capture the posterior mean and covariance accurate to the second order (9).

A. State-Space Model Formulation

The dynamic epidemiological system is represented as a non-linear discrete-time state-space model. Let $x_k \in R^L$ denote the augmented state vector at time step k , and $z_k \in R^M$ denote the observation vector (8). The system is governed by the following stochastic difference equations

$$\begin{aligned} x_k &= f(x_{k-1}, u_{k-1}) + w_{k-1} \\ z_k &= h(x_k) + v_k \end{aligned}$$

Where

- $f(\cdot)$: The non-linear transition function derived from the discretized UVWZ differential equations.

- $h(\cdot)$: The measurement function relating the state to the observed reported cases.
- w_k : The process noise vector, assumed to be zero-mean Gaussian white noise with covariance Q_k , i.e., $w_k \sim N(0, Q_k)$
- V_k : The measurement noise vector, assumed to be zero-mean Gaussian white noise with covariance R_k , i.e. $v_k \sim N(0, R_k)$

B. Unscented Transform

The core of the UKF is the Unscented Transform (UT), a method for calculating the statistics of a random variable which undergoes a non-linear transformation. The fundamental premise is that it is easier to approximate a probability distribution than it is to approximate an arbitrary non-linear function. Let x be a random variable with mean $\bar{x}$ and covariance P_x .

The UT propagates x through a non-linear function $y = g(x)$ by generating a set of deterministic sigma points. These points capture the true mean and covariance of x , and when propagated through $g(\cdot)$, they capture the posterior mean and covariance of y accurately to the second order (Taylor series expansion) for any non-linearity (9).

C. Sigma Point Construction

For an L -dimensional state vector x , a set of $2L + 1$ sigma points, denoted as X , is generated deterministically (9,11). The sigma points X_{k-1} is calculated as follows :

$$\begin{aligned} X_{0,k-1} &= x_{k-1} \\ X_{i,k-1} &= x_{k-1} + (\sqrt{(L + \lambda)P_{k-1}})_i, i = 1, \dots, L \\ X_{i,k-1} &= x_{k-1} - (\sqrt{(L + \lambda)P_{k-1}})_i, i = 1, \dots, 2L \end{aligned}$$

Where

- $(\sqrt{M})_i$ is the i -th column of the matrix square root of M
- $\lambda = \alpha^2(L + k) - L$ is a composite scaling parameter
- α determines the spread of the sigma points around the mean
- k is a secondary scaling parameter

D. Weight Calculation for Sigma Points

To reconstruct the statistics of the transformed distribution, specific weights are assigned to each sigma point (9,12). Separate weights are defined for the mean (W_m) and the covariance (w_c) :

$$\begin{aligned} W_m^{(0)} &= \frac{\lambda}{L + \lambda} \\ W_m^{(0)} &= \frac{\lambda}{L + \lambda} + (1 - \alpha^2 + \beta) \\ W_m^{(i)} &= W_c^{(i)} = \frac{1}{2(L + \lambda)}, i = 1, \dots, 2L \end{aligned}$$

Here, β is a parameter used to incorporate prior knowledge of the distribution of x , for gaussian distribution, $\beta = 2$ is optimal

E. Prediction Step

The prediction step projects the state and covariance estimates forward in time from step $k - 1$ to k using a non-linear system model.

E.1. State Prediction

The sigma points X_{k-1} are propagated through the non-linear transition function f (8,9) :

$$X_{i,k|k-1} = f(X_{i,k-1}, u_{k-1}), i = 0, \dots, 2L$$

The a priori state estimate is calculated as the weighted sum of the transformed sigma points:

$$x_{k|k-1} = \sum_{i=0}^{2L} W_m^{(i)} X_{i,k|k-1}$$

E.2. Covariance Prediction

The *a priori* error covariance is computed based on the deviation of the propagated sigma points from the predicted mean, adding the process noise covariance Q_k (9,11):

$$P_{k|k-1} = \sum_{i=0}^{2L} W_c^{(i)} [X_{i,k|k-1} - x_{k|k-1}] [X_{i,k|k-1} - x_{k|k-1}]^T + Q_k$$

F. Measurement Update Step

When a new measurement becomes z_k available, the filter corrects the predicted state (9).

F.1. Measurement Prediction

The propagated sigma points are transformed through the observation function h (7,9):

$$Y_{i,k|k-1} = h(X_{i,k|k-1}), i = 0, \dots, 2L$$

The predicted measurement mean is derived as:

$$z_{k|k-1} = \sum_{i=0}^{2L} W_m^{(i)} Y_{i,k|k-1}$$

F.2. Innovation Covariance

The measurement covariance (innovation covariance), denoted as S_k , accounts for the uncertainty in the measurement model and the measurement noise R_k (9):

$$S_k = \sum_{i=0}^{2L} W_c^{(i)} [Y_{i,k|k-1} - z_{k|k-1}] [Y_{i,k|k-1} - z_{k|k-1}]^T + R_k$$

F.3. Kalman Gain Computation

To compute the Kalman gain, the cross-covariance matrix $P_{x_k z_k}$ between the state and the measurement is required (9):

$$P_{x_k z_k} = \sum_{i=0}^{2L} W_c^{(i)} [X_{i,k|k-1} - x_{k|k-1}] [Y_{i,k|k-1} - z_{k|k-1}]^T$$

The Kalman gain K_k is then calculated as :

$$K_k = P_{x_k z_k} S_k^{-1}$$

F.4. State and Covariance Update

Finally, the state estimate and error covariance are updated using the actual measurement z_k (9) :

$$x_{k|k} = x_{k|k-1} + K_k (z_k - z_{k|k-1})$$

$$P_{k|k} = P_{k|k-1} - K_k S_k K_k^T$$

IV. RESULTS & DISCUSSION

A. Data Description

This study utilizes annual measles surveillance and vaccination coverage data to characterize long-term measles dynamics and to support state-space modeling using the Unscented Kalman Filter (UKF). The dataset spans the period 2004-2024 and integrates

epidemiological outcomes with immunization indicators, allowing the analysis to jointly consider disease incidence and population-level protection.

Table 2. Summary Statistics of Measles Incidence and Vaccination Coverage

Variable	Count	Mean	St. Dev	Min	25%	Median	75%	Max
Year	21	2014	6,2	2004	2009	2014	2019	2029
Measles_Reported_cases	21	12,86	7,823	494	6,962	15,099	18,869	29,171
WUENIC_Measles_containing_vaccine_1st_dose(%)	21	81,9	6,01	72	76	82	87	92
WUENIC_Measles_containing_vaccine_2nd_dose(%)	21	60,1	18.06	23	53	67	73	82

The dataset comprises 21 annual observations, with reported measles cases serving as the primary epidemiological outcome and vaccination coverage variables representing key immunization-related drivers of disease dynamics. In the Table 2, The measles incidence series exhibits substantial interannual variability, as reflected by the wide range and relatively large standard deviation, indicating the presence of outbreak years interspersed with periods of lower transmission. Such variability is characteristic of measles epidemiology and highlights the importance of modeling latent dynamics rather than relying solely on raw case counts.

Vaccination coverage for the first dose (MCV1) remains consistently higher and more stable than coverage for the second dose (MCV2), which shows markedly greater dispersion over the study period. This discrepancy suggests uneven completion of the measles immunization schedule, potentially resulting in cohorts with partial immunity that may sustain transmission. In the UKF-based state-space framework adopted in this study, reported measles cases are treated as noisy observations of an unobserved latent incidence process, while MCV1 and MCV2 coverage are incorporated as exogenous components influencing the hidden disease dynamics. This data structure is therefore well suited for disentangling underlying transmission patterns from observation noise and for assessing the role of vaccination coverage in shaping long-term measles incidence.

B. Exploratory Data Analysis (EDA)

EDA was conducted to examine the fundamental characteristics of the measles incidence and vaccination coverage data prior to state-space modeling. This step aims to identify temporal patterns, variability, and potential relationships among variables, which are essential for understanding the behavior of the observed system and for guiding the specification of the Unscented Kalman Filter (UKF) framework. By visually inspecting trends and pairwise associations, EDA provides initial evidence regarding nonlinearity, noise, and structural changes in the data that cannot be captured adequately by simple deterministic models.

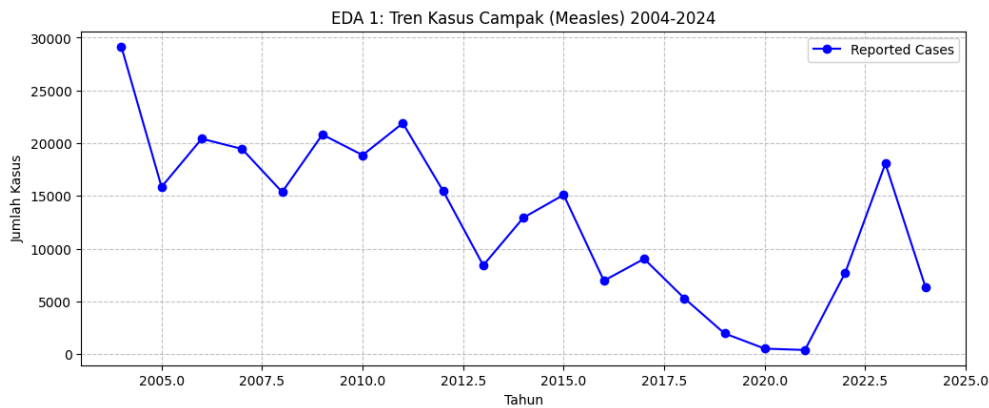


Figure 1. Temporal trend of reported measles cases from 2004 to 2024

Figure 1 illustrates the temporal evolution of reported measles cases over the period 2004-2024, revealing pronounced interannual fluctuations in measles incidence. Several high-incidence peaks are observed during the early and mid-2000s, followed by a gradual decline after 2015 that reaches very low reported levels around 2020-2021. This downward trend is subsequently interrupted by a sharp increase in cases during 2022-2023, indicating a resurgence of measles transmission after a period of suppression. The observed pattern suggests that measles incidence is governed by a combination of long-term trends and short-term shocks rather than a smooth monotonic process. Such dynamics are typical of infectious diseases influenced by changes in population immunity, surveillance practices, and external disruptions to health systems. The presence of abrupt increases and decreases in reported cases supports the need for a state-space approach, where observed case counts are treated as noisy measurements of an underlying latent incidence process that evolves dynamically over time.

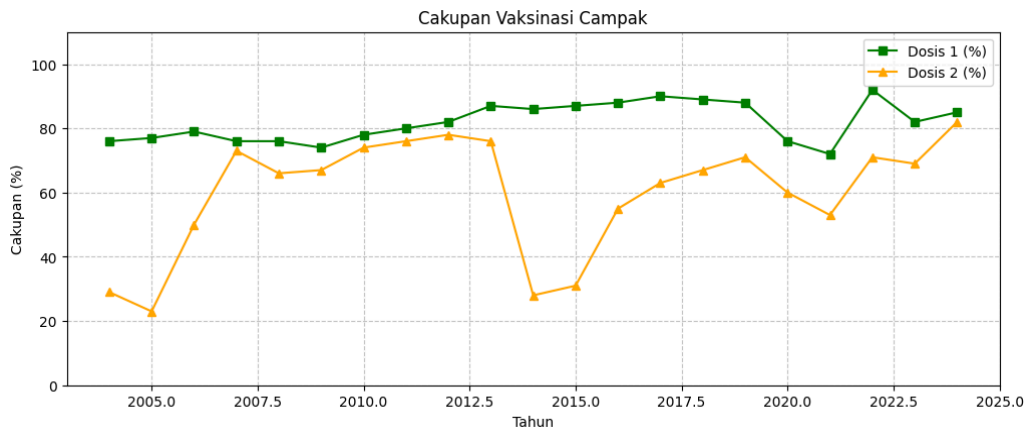


Figure 2. Annual coverage of measles-containing vaccine for the first dose (MCV1) and second dose (MCV2)

Figure 2 presents the annual coverage of measles-containing vaccine for the first dose (MCV1) and second dose (MCV2) throughout the study period. First-dose coverage remains relatively high and stable, generally exceeding 75% and reaching above 90% in several recent years. In contrast, second-dose coverage exhibits substantially greater variability, including sharp declines around 2014-2015 followed by a gradual recovery in subsequent years. This divergence between MCV1 and MCV2 coverage indicates incomplete vaccination among portions of the population, which may leave immunity gaps that sustain measles transmission. The higher volatility observed in MCV2 coverage suggests that completion of the full vaccination schedule is more sensitive to programmatic and systemic disruptions than initial vaccine uptake. From a modeling perspective, these vaccination series provide important exogenous information that may influence the latent transmission dynamics of measles and

are therefore incorporated into the state-space model to account for time-varying population protection.

Pairwise Scatter Plots of Measles Cases and Vaccine Coverage

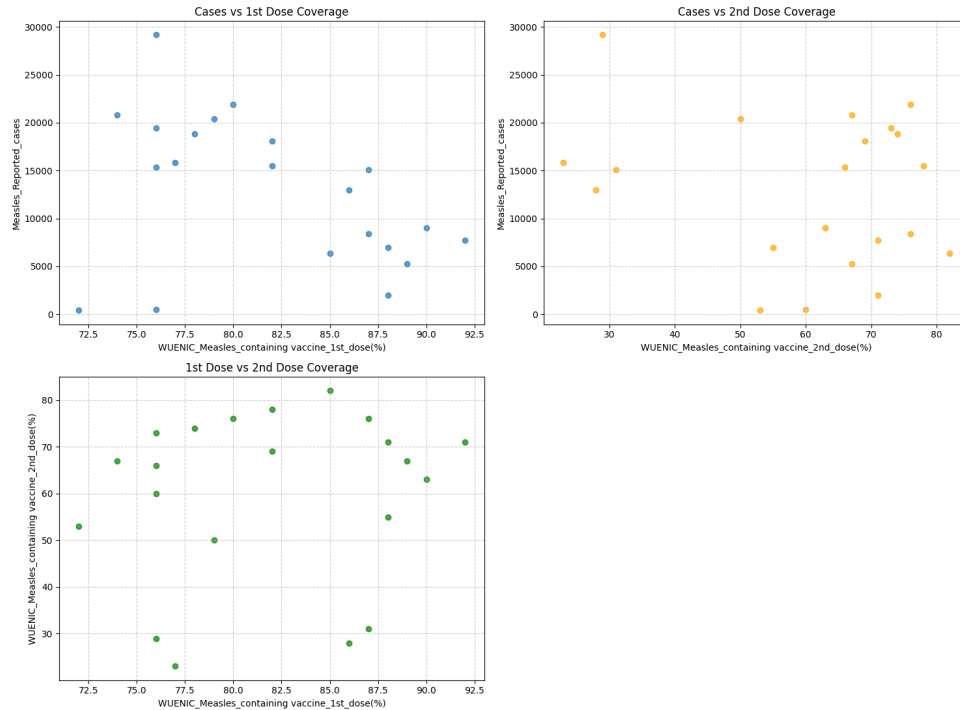


Figure 3. Pairwise scatter plots of reported measles cases and vaccination coverage variables

Figure 3 shows pairwise scatter plots examining the relationships between reported measles cases and vaccination coverage for the first and second doses, as well as the association between MCV1 and MCV2 coverage. The scatter plot of measles cases versus first-dose coverage indicates a generally negative relationship, where higher MCV1 coverage tends to be associated with lower case counts, although considerable dispersion is evident. This suggests that while first-dose vaccination contributes to reducing transmission, it alone is insufficient to fully prevent outbreaks. A similar but more scattered relationship is observed between measles cases and second-dose coverage, implying that low MCV2 coverage can coincide with both high and low incidence years due to the influence of additional unobserved factors. The relationship between MCV1 and MCV2 coverage is positive but not strictly linear, indicating that improvements in first-dose delivery do not always translate proportionally into higher second-dose completion. Overall, these patterns highlight nonlinear interactions and latent effects within the data, reinforcing the suitability of a UKF-based state-space framework for capturing the complex dynamics of measles transmission.

C. State-Space Model Specification

The measles surveillance data analyzed in this study were modeled using a state-space representation to distinguish between the underlying epidemiological process and the reported case observations. This approach is particularly suitable for infectious disease data, where reported cases often contain substantial noise due to reporting delays, underreporting, and changes in surveillance intensity. Based on the implemented notebook, the state-space model was constructed to capture both the temporal evolution of measles incidence and the influence of vaccination coverage as exogenous factors.

In this study, the state vector consists of four components that jointly describe the latent dynamics of measles incidence. The primary component represents the latent incidence level, which corresponds to the true underlying measles burden in the population. To improve numerical stability and reduce the impact of extreme values, reported measles cases were transformed using a logarithmic transformation before being assimilated into the

model. This transformation enables the state-space model to handle the highly skewed distribution commonly observed in epidemiological count data.

The second state component represents the temporal trend, which captures the long-term tendency of measles incidence to increase or decrease over time. This component allows the model to separate systematic changes in disease dynamics from short-term fluctuations caused by noise or irregular reporting. In addition, two further state components were introduced to represent the effects of vaccination coverage, specifically the first-dose and second-dose measles-containing vaccines. These components allow the model to explicitly incorporate immunization coverage as exogenous drivers that may influence the evolution of measles incidence.

The observation process links the latent state to the reported measles cases. In the implemented model, the observed data consist solely of reported measles cases, which are treated as noisy measurements of the latent incidence level. Importantly, only the incidence level is directly observed, while the trend and vaccination effect parameters remain unobserved and are inferred indirectly through the filtering process. This design reflects the realistic epidemiological assumption that structural drivers of disease dynamics cannot be directly measured from routine surveillance data.

Overall, the specified state-space model provides a structured and interpretable framework for separating signal from noise in measles surveillance data. By combining latent incidence dynamics, long-term trends, and vaccination effects within a unified representation, the model establishes a robust foundation for applying the Unscented Kalman Filter to estimate and smooth measles incidence over the study period from 2004 to 2024.

Table 3. Definition of State-Space Model Components.

Component	Description	Role in the Model
Latent incidence level	Log-transformed underlying measles incidence	Represents the true disease intensity
Trend	Long-term temporal change in incidence	Captures systematic increase or decrease over time
Vaccine effect (1st dose)	Effect of first-dose measles vaccine coverage	Models short- to medium-term preventive impact
Vaccine effect (2nd dose)	Effect of second-dose measles vaccine coverage	Represents sustained immunization influence
Observed measles cases	Reported measles cases (log-transformed)	Noisy measurement of the latent incidence level

Table 3 summarizes the components of the state-space model used in this study. The table highlights the distinction between latent states, which represent unobserved epidemiological processes, and observed variables, which correspond to reported measles cases. The inclusion of vaccination coverage effects as separate state components enables the model to account for external public health interventions that influence disease dynamics. By structuring the model in this way, the state-space formulation supports both interpretability and flexibility, allowing the Unscented Kalman Filter to effectively estimate the underlying measles incidence while accounting for uncertainty and measurement noise.

D. Unscented Kalman Filter Implementation

The Unscented Kalman Filter (UKF) was implemented to estimate the latent dynamics of measles incidence using the state-space model described in the previous section. The implementation follows a recursive filtering procedure in which the system state is iteratively predicted and updated for each annual observation. The UKF framework allows the incorporation of non-linear system dynamics and exogenous vaccination inputs while maintaining numerical stability. The implementation was carried out using a custom UKF class, ensuring full control over parameter selection, state evolution, and uncertainty propagation throughout the filtering process.

D.1. UKF Parameter Initialization

The performance of the Unscented Kalman Filter is strongly influenced by the choice of its initial parameters. In this study, the UKF was initialized using information derived from the measles surveillance data and reasonable assumptions regarding uncertainty in the underlying epidemiological process. The state vector was defined with four components, representing the latent incidence level, temporal trend, and the effects of first-dose and second-dose vaccination coverage. The initial value of the latent incidence level was obtained from the first observed data point after logarithmic transformation, while the initial trend was approximated using the difference between the first two observations. The vaccination effect parameters were initialized with neutral values, reflecting the absence of strong prior assumptions regarding their magnitude at the beginning of the study period.

The initial state covariance matrix was specified as a diagonal matrix with relatively larger variances for the incidence level and trend, indicating higher initial uncertainty in these components. Smaller variances were assigned to the vaccination effect parameters to prevent abrupt and unrealistic parameter changes during the early filtering stages. Process noise covariance was configured to allow gradual evolution of the state variables over time, ensuring that the filter remains responsive to long-term structural changes without overfitting short-term fluctuations. Measurement noise was set to reflect uncertainty in reported measles cases, acknowledging potential reporting errors and inconsistencies in surveillance data.

In addition, the Unscented Kalman Filter tuning parameters were selected to ensure numerical stability and accurate uncertainty representation. A small value of the spread parameter was used to limit excessive dispersion of sigma points, while the Gaussian prior parameter was set to its optimal value for normally distributed noise. Together, these initialization choices provide a balanced configuration that supports stable filtering, reliable uncertainty propagation, and meaningful interpretation of the estimated measles dynamics throughout the study period.

Table 4. UKF Parameter Initialization.

Parameter	Value / Configuration	Description
State dimension	4	Latent incidence level, trend, first-dose and second-dose vaccine effects
Initial state	Derived from first observation	Starting condition of the latent state
Initial covariance matrix	Diagonal matrix	Represents initial uncertainty in each state component
Process noise covariance	Tuned diagonal matrix	Controls smoothness and adaptability of state evolution
Measurement noise covariance	Scalar matrix	Represents uncertainty in reported measles cases
Alpha (α)	1×10^{-3}	Controls the spread of sigma points
Beta (β)	2	Optimal for Gaussian-distributed noise
Kappa (κ)	0	Scaling parameter for sigma point generation

Table 4 summarizes the initialization settings used in the UKF implementation. The table highlights how uncertainty is explicitly controlled through covariance specification and tuning parameters, ensuring that the filter can effectively balance model-based predictions with information from observed data. The chosen parameter configuration reflects a conservative initialization strategy, allowing the filter to gradually learn the underlying measles dynamics rather than imposing strong prior assumptions at the start of the filtering process.

D.2. Sigma Point Generation

After the initialization stage, the Unscented Kalman Filter represents the uncertainty of the current state estimate through a deterministic set of sigma points. In this study, sigma points were generated at each time step based on the current state mean and covariance

matrix obtained from the previous filtering iteration. These sigma points provide a compact yet informative representation of the state distribution, allowing the filter to capture potential non-linear effects without relying on linear approximations.

The number of sigma points was determined by the dimensionality of the state vector, resulting in a symmetric set of points distributed around the current state estimate. Each sigma point reflects a plausible realization of the latent measles dynamics, including variations in the incidence level, trend, and vaccination effect parameters. The spread of the sigma points was controlled by the selected UKF tuning parameters, ensuring that the points adequately represent uncertainty while avoiding excessive dispersion that could lead to numerical instability.

By propagating these sigma points through the state transition model, the UKF preserves both the central tendency and the uncertainty structure of the system. This approach enables the filter to accurately account for the non-linear influence of vaccination coverage on measles incidence dynamics. As a result, sigma point generation plays a crucial role in maintaining the robustness and stability of the filtering process, forming the basis for reliable prediction and update steps in subsequent stages of the UKF implementation.

D.3. Prediction Step

In the prediction step, the Unscented Kalman Filter propagates the previously generated sigma points forward in time to obtain a prior estimate of the system state before incorporating the current observation. In this study, each sigma point was advanced using the specified state transition mechanism, which captures the evolution of measles incidence as a function of its past level, long-term trend, and the influence of vaccination coverage. This step produces a predicted representation of the latent measles dynamics for the current year.

The prediction process accounts for uncertainty growth by incorporating process noise, reflecting unobserved disturbances and natural variability in epidemiological dynamics. As a result, the predicted state distribution exhibits increased uncertainty compared to the previous time step, particularly during periods of rapid change in reported cases. This behavior is consistent with the expectation that uncertainty accumulates when new observations have not yet been assimilated.

Overall, the prediction step provides a prior estimate that summarizes the expected evolution of measles incidence based solely on historical information and vaccination inputs. This predicted state serves as the basis for subsequent correction during the measurement update stage, ensuring that the filter balances model-driven expectations with information derived from observed surveillance data.

D.4. Measurement Update Step

The measurement update step integrates the observed measles surveillance data into the predicted state obtained from the previous prediction step. In this study, the update process was designed to reflect the realistic structure of epidemiological data, where only reported measles cases are directly observed, while other state components remain latent. Consequently, the UKF updates only the latent incidence level using the current observation, whereas the trend and vaccination effect parameters are adjusted indirectly through their covariance relationships with the observed state.

During this step, the discrepancy between the predicted incidence level and the observed measles cases is used to correct the prior state estimate. This correction reduces uncertainty in the latent incidence level and improves alignment with the surveillance data, while preserving smooth and stable estimates of unobserved parameters. By selectively updating only the observable component of the state vector, the filter avoids overfitting and maintains epidemiological interpretability.

The measurement update step therefore plays a critical role in balancing model-based predictions and real-world observations. It ensures that the UKF remains responsive to actual reported cases while still leveraging the underlying state-space structure to infer long-term trends and vaccination effects that cannot be directly measured from the data.

Table 5. Measurement Update Configuration in the UKF Implementation.

Aspect	Configuration in This Study	Interpretation
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Observed variable	Reported measles cases (log-transformed)	Direct measurement from surveillance data
Updated state component	Latent incidence level	State directly informed by observations
Unobserved state components	Trend, vaccine effect (1st dose), vaccine effect (2nd dose)	Estimated indirectly through covariance structure
Measurement frequency	Annual	Consistent with data resolution
Measurement noise	Scalar covariance	Represents reporting and surveillance uncertainty
Update strategy	Partial state update	Prevents unrealistic parameter fluctuations

Table 5 summarizes how measurement information is incorporated into the UKF during the update step. The table highlights that only the latent incidence level is directly corrected using observed data, while other state components are refined indirectly. This selective update strategy ensures numerical stability and prevents abrupt changes in structural parameters such as trend and vaccination effects. By explicitly modeling measurement uncertainty, the UKF accounts for imperfections in surveillance data and produces more reliable estimates of the underlying measles dynamics

E. Filtering Results

This section presents the results of the filtering process obtained using the Unscented Kalman Filter applied to measles surveillance data. The focus of this section is to evaluate how effectively the UKF estimates the latent measles incidence by comparing the filtered state with observed data and by examining the associated uncertainty. The results highlight the ability of the UKF to smooth noisy observations while preserving meaningful temporal patterns in measles dynamics over the study period from 2004 to 2024.

E.1. Estimated State vs Observed Data

The comparison between the estimated latent state and the observed measles cases demonstrates that the UKF successfully captures the underlying dynamics of measles incidence. The filtered estimates follow the overall trajectory of reported cases while exhibiting a smoother temporal profile. Sharp fluctuations and irregular spikes present in the raw surveillance data are substantially reduced in the filtered state, indicating effective noise attenuation by the UKF.

During periods of rapid change in reported cases, the filtered estimates respond gradually rather than abruptly, reflecting the influence of the state-space structure and process noise constraints. This behavior suggests that the UKF prioritizes consistency in the latent disease dynamics over short-term reporting variability. Overall, the filtered state provides a more stable and interpretable representation of measles incidence compared to the raw observations, making it suitable for analyzing long-term trends and structural influences such as vaccination coverage.

E.2. Uncertainty Analysis

In addition to providing point estimates of measles incidence, the Unscented Kalman Filter quantifies the uncertainty associated with each filtered state estimate. This uncertainty reflects the confidence of the model in representing the latent measles dynamics and evolves over time as new observations are assimilated. In the implemented UKF, uncertainty is represented by the state covariance matrix, which is updated recursively throughout the filtering process.

The results indicate that uncertainty varies across the study period, responding to changes in the variability of reported measles cases. Periods characterized by relatively stable reported cases exhibit lower uncertainty, suggesting higher confidence in the estimated latent incidence. Conversely, during periods with sharp increases or irregular fluctuations in reported cases, the uncertainty of the filtered estimates increases. This

behavior demonstrates that the UKF appropriately accounts for heightened unpredictability in the surveillance data rather than forcing overly confident estimates.

Furthermore, the gradual evolution of uncertainty over time reflects the balance achieved by the UKF between model-based expectations and observation-driven corrections. By explicitly propagating uncertainty through the prediction step and reducing it during the measurement update step, the filter avoids abrupt changes in confidence and maintains stable estimation behavior. This adaptive uncertainty representation is particularly important in epidemiological applications, where data quality and reporting consistency may vary substantially over time.

Table 6. Summary of Filtering Uncertainty for the Latent Measles Incidence

Uncertainty Indicator	Value	Interpretation
Mean variance	0.381	Average uncertainty of the filtered incidence level
Minimum variance	0.365	Periods of highest estimation confidence
Maximum variance	0.398	Periods with highest uncertainty
Temporal pattern of uncertainty	Time-varying	Uncertainty adapts to data variability

Based on Tabel 6, the uncertainty analysis indicates that the Unscented Kalman Filter provides relatively stable estimates of the latent measles incidence throughout the study period. The mean variance of 0.381 suggests a moderate level of overall uncertainty in the filtered incidence estimates, reflecting a balanced confidence between model assumptions and observed data. The small difference between the minimum variance (0.365) and maximum variance (0.398) indicates that the uncertainty does not fluctuate excessively over time, implying that the filter maintains consistent estimation performance despite variations in reported cases. The time-varying pattern of uncertainty further demonstrates the adaptive nature of the UKF, as the model dynamically adjusts its confidence in response to changes in data variability rather than assuming constant reliability across all periods.

F. Model Performance Evaluation

Model performance was assessed by comparing the UKF-filtered estimates against the observed annual measles case counts using RMSE, NRMSE, and MAPE. The following model performance can be seen in Table 7 .

Table 7. Performance Result

Metrics	Value
RMSE	763.79
NRMSE	0.0265
MAPE	5.41%

Based on Table 7, the UKF achieved an RMSE of 763.79 cases, indicating that, on average, the filtered estimates deviate from the observed surveillance counts by fewer than one thousand cases per year. This level of error is relatively small when compared to the overall magnitude of measles cases in the dataset (which ranges from hundreds to tens of thousands), suggesting that the UKF provides a close fit to the reported incidence series while still performing smoothing. As shown in Figure 4, the UKF filtered curve closely overlaps with the observed case trajectory across 2004-2024, while producing a smoother profile that reduces irregular year-to-year fluctuations and highlights the underlying incidence pattern.

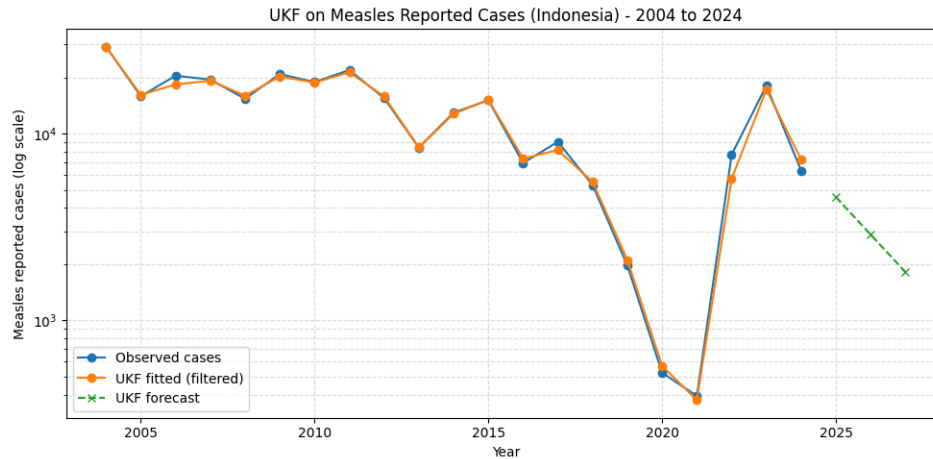


Figure 4. Comparison between observed measles cases, UKF-filtered estimates, and UKF-based forecasts

To provide a scale-independent interpretation, the NRMSE was 0.0265, meaning the estimation error is approximately 2.65% of the observed data range. This low normalized error indicates strong agreement between the UKF output and the observed time series across the full study window. Consistent with this finding, the MAPE was 5.41%, implying that the UKF estimates differ from observed values by about 5% on average in relative terms, which supports the conclusion that the filtering approach yields high-accuracy reconstruction of measles incidence at an annual resolution. Figure 4 further indicates that the UKF successfully follows major structural changes in the series, including the low-incidence period around 2020-2021 and the subsequent resurgence in 2022-2023, demonstrating responsiveness to regime shifts without excessive sensitivity to noise. Overall, the metric results demonstrate that the UKF successfully balances model-based prediction and observation-based correction, producing estimates that closely track reported measles cases while attenuating short-term irregularities that are likely driven by measurement noise, reporting variability, and surveillance inconsistencies. In addition, Figure 1 visualizes the UKF forecast beyond the observed window, providing an interpretable projection of near-term measles dynamics derived from the estimated latent state and its associated uncertainty structure.

G. Discussion

The filtering results indicate that the Unscented Kalman Filter effectively captures the underlying dynamics of measles incidence in Indonesia over 2004-2024. Visual comparison between observed and filtered series shows that the UKF follows the main trajectory of the reported cases across the full period, including the long decline toward 2020-2021 and the subsequent resurgence around 2022-2023, while producing a smoother representation of the incidence process. This behavior is consistent with the purpose of state-space filtering: separating latent epidemiological signals from observation noise without losing the structural features of the time series.

A key strength of the UKF in this context is its suitability for nonlinear systems and noisy data. Unlike EKF, which relies on local linearization, the UKF uses sigma points to propagate uncertainty through nonlinear dynamics, which can be advantageous when disease incidence evolves in a non-stationary way due to changing immunity, interventions, or behavioral patterns. The close match between filtered estimates and observations, combined with low NRMSE and MAPE, suggests that the UKF implementation provides stable and accurate estimation even across years with rapid changes in reported cases.

From the parameter interpretation output, the final filtered state indicates a positive trend component ($\text{trend} = 0.427$), implying an increasing tendency in the latent log-incidence level toward the end of the observation window. This is epidemiologically plausible given the resurgence visible in the most recent years of the dataset. The estimated vaccination effect parameters show $b_1 > 0$ and $b_2 < 0$ in the final state, where a negative coefficient is typically interpreted as higher vaccine coverage being associated with lower measles incidence (on the log scale). The negative b_2 aligns with expectations that improved second-dose coverage

contributes to reducing transmission by strengthening population immunity. The positive b_1 at the final time step may reflect confounding dynamics in the annual data, such as delayed outbreak responses, shifts in surveillance intensity, or the fact that MCV1 coverage alone may not capture effective immunity if MCV2 completion lags, highlighting that vaccination effects can vary over time and may not be fully identifiable from annual aggregates.

The uncertainty analysis supports the stability of the filtering process, with variance values remaining within a narrow band (0.365-0.398). This relatively consistent uncertainty suggests that the UKF maintained stable confidence in the latent incidence estimation across years, rather than becoming overly uncertain during outbreaks or overly confident during low-incidence periods. Such behavior is desirable for surveillance modeling, as it indicates the filter is robust to fluctuations while still adapting to changes in the observations. Nevertheless, several limitations should be acknowledged. First, the use of annual data reduces temporal resolution and can mask within-year seasonality, which is important in measles transmission. Second, reported cases may be affected by underreporting and changes in surveillance effort, meaning that the observation process may not represent true incidence uniformly over time. Third, vaccination coverage (MCV1 and MCV2) is a population-level indicator and does not directly measure immunity distribution across age cohorts, which may influence outbreaks even when average coverage appears high. Future work could extend this framework to monthly surveillance data (if available), incorporate additional covariates (e.g., birth rates, mobility, supplementary immunization activities), and test time-varying vaccination effects explicitly to improve interpretability of immunization impacts.

In summary, the UKF-based state-space model demonstrates strong performance for estimating and smoothing measles incidence in Indonesia across 2004-2024, achieving low error metrics and producing interpretable latent dynamics. The results highlight the value of filtering approaches for infectious disease surveillance, especially when data contain noise and structural changes that challenge classical deterministic modeling.

V. CONCLUSIONS

This study demonstrated that the application of the Unscented Kalman Filter (UKF) within a state-space modeling framework provided accurate and stable estimates of measles incidence dynamics in Indonesia over the period 2004-2024. By modeling reported measles cases as noisy observations of a latent epidemiological process, the UKF effectively separated the underlying disease signal from annual reporting fluctuations, allowing a clearer representation of long-term transmission patterns.

The filtering results showed that the UKF successfully captured major structural changes in measles incidence, including the pronounced decline around 2020-2021 and the subsequent resurgence in later years. Model performance metrics indicated a strong level of accuracy, with relatively low estimation errors (RMSE of 763.79 cases, NRMSE of 0.0265, and MAPE of 5.41%), confirming the suitability of the UKF for annual measles surveillance data. These results suggest that the proposed approach can reliably smooth irregular observations while preserving meaningful epidemiological trends.

Parameter interpretation revealed a positive latent trend toward the end of the observation period, which is consistent with the observed resurgence of measles cases. In addition, the estimated vaccination effect parameters indicated a more consistent protective influence of second-dose vaccine coverage compared to first-dose coverage. This finding highlights the importance of completing the full measles immunization schedule to achieve sustained population immunity and reduce transmission risk.

The uncertainty analysis further indicated that the UKF adaptively adjusted its confidence in response to data variability, producing robust and epidemiologically plausible estimates across different transmission regimes. However, several limitations remain, including the use of annual data that do not capture seasonal patterns, the potential presence of underreporting in surveillance records, and the reliance on aggregated vaccination coverage indicators that do not fully represent immunity distribution within the population.

Future research should consider applying the proposed framework to data with higher temporal resolution, incorporating additional epidemiological covariates such as birth rates, population mobility, or supplementary immunization activities, and exploring fully

time-varying or hybrid modeling approaches. Overall, this study confirms that the Unscented Kalman Filter represents a promising and relevant tool for infectious disease surveillance analysis, particularly in settings characterized by non-linear, non-stationary, and uncertainty-prone epidemiological data such as measles incidence in Indonesia.

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FINANCING

The authors did not receive financing for the development of this research.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

AUTHORSHIP CONTRIBUTION:

Conceptualization : Muhammad Muafi, Andreas Hendra Herwanto, Herman Effendi, Pradipta Deska Pryanda.

Data curation: Andreas Hendra Herwanto

Formal analysis: Herman Effendi, Andreas Hendra Herwanto



Research: Muhammad Muafi

Methodology: Herman Effendi, Andreas Hendra Herwanto

Project management: Andreas Hendra Herwanto

Resources: Muhammad Muafi

Software: Herman Effendi

Supervision: Herman Effendi

Validation: Muhammad Muafi

Display: Andreas Hendra Herwanto

Drafting - original draft: Muhammad Muafi

Writing - proofreading and editing: Muhammad Muafi, Andreas Hendra Herwanto